

Chemistry, 2015 EC, entrance exam, version 1

Grade 12 · Chemistry · 2015 E.C.

Time allowed: 120 minutes

Questions: 75

Total marks: 75

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following is an intensive physical property of matter? (1)

A. Mass	B. Volume
C. Length	D. Density

2. All of the following have four significant figures except (1)

A. 0.05040	B. 0.0012
C. 09.060	D. 1.005

3. Which of the following is correct? (1)

A. 1 L = 1 dm ³	B. 1 L = 10 dm ³
C. 10 L = 1 dm ³	D. 1 L = 1 m ³

4. What is the first step of the scientific method? (1)

A. Performing an experiment	B. Forming a hypothesis
C. Making observation	D. Predicting the result of an experiment

5. The equivalent temperature value in Fahrenheit scale for 50°C is (1)

A. 122	B. 50
C. 100	D. 90

6. Which of the following is possible while doing an experiment? (1)

A. Washing hand with soap	B. Smelling chemicals
C. Dangling hair	D. Chewing gum

7. According to Modern atomic theory, one of the Dalton's atomic theory was wrong? (1)

A. Atoms combine in small whole number ratios to form compounds
B. Atoms of the same element are identical in mass and other properties.
C. Elements are made up of small particles called atoms
D. Atoms of one element differ from atoms of other element.

8. The main purpose of Robert Millikan's oil drop experiment is to determine (1)

A. the speed of electron
B. the ratio of mass of electron to charge of electron
C. the mass of electron
D. the magnitude of charge of electron

9. Which region of electromagnetic spectrum is capable of inducing electron transition with the greatest energy? (1)

A. Orange	B. Red
C. Violet	D. Green

10. What is the wavelength of the yellow sodium emission, which has a frequency of $5.09 \times 10^{14} \text{ s}^{-1}$? ($c = 3 \times 10^8 \text{ m/s}$) (1)

A. 400 nm
B. 589 nm
C. 670 nm
D. 890 nm

11. What is the ratio of the energy of a photon of 300 nm wavelength radiation to that of 600 nm radiation? (1)

A. 1:2
B. 1:1
C. 3:1
D. 2:1

12. The ejection of electron from metals by shining light on the metal surface is (1)

A. work function
B. binding energy
C. photoelectric effect
D. speed of light

13. The subshell representation for $n = 4, l = 3$ is (1)

A. 4p
B. 4d
C. 4s
D. 4f

14. What are the possible values of the azimuthal quantum number for an electron in the 4th energy level? (1)

A. 0,1,2,3
B. 1,2,3
C. 2,3,4
D. 3,4

15. How many 3d electrons are present in the ground state of $^{52}_{24}\text{Cr}$? (1)

A. 4
B. 6
C. 5
D. 10

16. Within a specified period, an increase in atomic number is usually accompanied by (1)

A. an increase in atomic radius and an increase in electronegativity
B. a decrease in atomic radius and an increase in electronegativity
C. an increase in atomic radius and a decrease in electronegativity
D. a decrease in atomic radius and a decrease in electronegativity

17. In which of the following block of the periodic table is an element with atomic number of 20 placed? (1)

A. s-block
B. p-block
C. d-block
D. f-block

18. A substance that does NOT conduct electricity as a solid but does conduct electricity when melted is most likely classified as: (1)

A. a metal
B. a molecular compound
C. an ionic compound
D. a non-metal

19. Which of the following ionic compounds has the highest lattice energy? (1)

A. NaCl
B. NaF
C. MgO
D. AlN

20. Which one of the substance contains ionic, covalent and dative bonds between their atoms? (1)

A. NH_4CN
B. KI
C. Mg_3N_2
D. CHOH

32. For a certain reaction, rate = $k[A]^{3/2}$. If the rate is 0.020 M/s when $[A] = 1.0$ M, what is the rate when $[A] = 0.64$ M? (1)
- A. 0.0033 M/s
B. 0.04 M/s
C. 0.033 M/s
D. 0.01024 M/s
33. Substances that inhibit a reaction are: (1)
- A. promoters
B. negative Catalysts
C. positive Catalysts
D. enzymes
34. If we increase the concentration of a reactant, what happens to the collisions between particles? (1)
- A. There are fewer collisions.
B. There are the same numbers of collisions but they have less energy.
C. There are more collisions.
D. There are the same numbers of collisions but they have more energy.
35. Which of the following rate law is third order overall? (1)
- A. Rate = $k[A][B]^3$
B. Rate = $k[A][B]^2$
C. Rate = $k[A]^3[B]^3$
D. Rate = $k[A]^5[B]^2$
36. The reaction between NO and I_2 is second order in NO and first order in I_2 . What change occurs in the rate of reaction if the concentration of each reactant is tripled? (1)
- A. 18 fold increase
B. 9 fold increase
C. 27 fold increase
D. 3 fold increase
37. Ozone (O_3) is depleted from tropospheric atmosphere by the following cycle: Step1: $NO + O_3 \rightarrow NO_2 + O_2$; Step2: $NO_2 + O \rightarrow 2O_2$; Net: $O_3 + O \rightarrow 2O_2$. What observation could you make about the cycle? (1)
- A. NO_2 is a catalyst
B. O is a catalyst
C. O is an intermediate
D. NO_2 is an intermediate
38. Suppose reactions $A \rightarrow B$ and $B \rightarrow A$ are both elementary processes with rate constants of 800 s^{-1} and 4000 s^{-1} respectively. What is the value of the equilibrium constant for $A \rightleftharpoons B$? (1)
- A. 0.2
B. 20
C. 0.02
D. 2
39. Considering the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $\Delta H = +180\text{ kJ/mol}$, the production of NO is increased by: (1)
- A. increasing temperature
B. increasing pressure
C. decreasing temperature
D. decreasing pressure
40. In the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $K = 100$. What is $[O_2]$ if $[SO_2] = [SO_3]$? (1)
- A. 0.001
B. 0.1
C. 100
D. 0.01
41. The relation between the equilibrium constant in terms of concentration and pressure for the equilibrium reaction $2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + H_2O(g) + CO_2(g)$ is (1)
- A. $K_p = K_cRT$
B. $K_c = K_p(RT)^2$
C. $K_p = K_c(RT)^2$
D. $K_c = K_pRT$

42. Which one of the following indicates the correct expression for the equilibrium constant (K) for $\text{PCl}_5(\text{s}) \rightarrow \text{PCl}_3(\text{l}) + \text{Cl}_2(\text{g})$?

- A. $[\text{PCl}_3][\text{Cl}_2]/[\text{PCl}_5]$ B. $[\text{Cl}_2]$
C. $[\text{PCl}_3][\text{Cl}_2]$ D. $[\text{PCl}_5]/[\text{PCl}_3][\text{Cl}_2]$

43. Which of the following statement correctly describes a chemical reaction at equilibrium? (1)

- A. The concentrations of the products and reactants are equal
- B. The rate of the forward reaction is less than the reverse reaction
- C. The rate of the forward reaction is equal the reverse reaction
- D. The rate of the forward reaction is greater than the reverse reaction

44. The number of phase in a mixture of sodium chloride and water is (1)

- A. 1 B. 2
C. 3 D. 4

45. What is the chemical name for Aspirin? (1)

- A. Sodium salicylate
B. Salicylic acid
C. Methyl salicylate
D. Acetyl salicylic acid

46. Which compound is a carboxylic acid? (1)

- A. $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$ B. $\text{CH}_3\text{CH}_2\text{COOH}$
C. $\text{CH}_3\text{CH}_2\text{CHO}$ D. $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$

47. Hydrolysis of propyl butanoate produces (1)

- A.** Propanol and butanoic acid **B.** Butanol and propanoic acid
C. Butanol and Propanoic acid **D.** Ethanol and Butanoic acid

48. What is the process that converts liquid vegetable oils to solid fats? (1)

- A.** Hydration **B.** Hydrolysis
C. Hydrogenation **D.** Saponification

49. Which of the following statements concerning fats and oils is incorrect? (1)

- A.** They are also called triacyl glycerols
- B.** They are also called triglycerides
- C.** They are fatty acids salts
- D.** They are glycerol triesters

50. Which of the following types of compounds are expected products from the saponification of a fat? (1)

- A.** Glycerol, fatty acids and fatty acid salts
B. Fatty acid salts and fatty acid
C. Glycerol and fatty acids
D. Glycerol and fatty acid salts

51. Which of the following is NOT a solution? (1)

- A.** Milk **B.** Brass
C. Whisky **D.** Coca cola drink

52. The process of solute particles being surrounded by solvent molecules is (1)

- A. saturation
B. solvation
C. dehydration
D. suspension

53. One of the following does not speed up the rate of dissolution? (1)
- A. Increasing temperature
B. Increasing pressure of gas
C. Increasing surface area
D. Decreasing surface area
-
54. When a crystal solute is added to its supersaturated solution, the solute crystal will (1)
- A. slightly dissolve
B. remain unchanged
C. grow bigger
D. dissolve completely
-
55. The solubility of oxygen gas in water at 25°C and 1 atm pressure is 0.041 g/L. What is the solubility of oxygen in water at 25°C and 3 atm? (1)
- A. 0.123
B. 0.31
C. 0.041
D. 0.014
-
56. Calculate the mole fraction of H₂O and CH₃CH₂OH in a mixture of equal masses of H₂O & CH₃CH₂OH. (H=1, O=16) (1)
- A. 0.9 & 0.1
B. 0.72 & 0.28
C. 0.5 & 0.5
D. 0.6 & 0.4
-
57. The molarity of nitrate ion in a solution that contains 0.9 M Al(NO₃)₃ is (1)
- A. 0.3 M
B. 2.7 M
C. 0.9 M
D. 1.8 M
-
58. What is the normality of 0.2 M solution of Ca₃(PO₄)₂? (1)
- A. 0.2 N
B. 3 N
C. 1.2 N
D. 0.6 N
-
59. What is the volume of water required to dilute 40 mL of 2 M HCl to 0.5 M HCl? (1)
- A. 80 mL
B. 40 mL
C. 160 mL
D. 120 mL
-
60. What volume of 2 M H₂SO₄ is needed to react exactly with 240 g of magnesium? (Mg=24, S=32, O=16, H=1) (1)
- A. 5 L
B. 10 L
C. 2.5 L
D. 2 L
-
61. The spectator ions are: 2AgNO₃(aq) + Na₂CrO₄(aq) → Ag₂CrO₄(s) + 2NaNO₃(aq) (1)
- A. Na⁺, NO₃⁻
B. Ag⁺, NO₃⁻
C. Na⁺, CrO₄²⁻
D. Ag⁺, CrO₄²⁻
-
62. The boiling point of pure ethanol is 78.5°C. What is the boiling point of a solution when 3.4 g of vanillin is dissolved in ethanol? (1)
- A. Less than 78.5°C
B. Equals 78.5°C
C. Greater than 78.5°C
D. Equals -78.5°C
-
63. A solution is prepared by mixing 1 mole of ethanol and 2 mole of water at 25°C. What is the total vapour pressure above the solution? (vapour pressure of pure ethanol = 44 torr, pure water = 23.8 torr) (1)
- A. 33.9 torr
B. 30.5 torr
C. 45 torr
D. 67.8 torr

64. The conjugate base of H_3O^+ is (1)
- A. OH^- B. H^+
C. NH D. H_2O
-
65. Given the reaction: $\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{HPO}_4^{2-}$. Which of the following represents a conjugate acid-base pair? (1)
- A. H_2PO_4^- and H_2O B. H_2PO_4^- and HPO_4^{2-}
C. H_2PO_4^- and H_3O^+ D. H_2O and HPO_4^{2-}
-
66. The pH value of three acids A, B and C are 1, 2 and 3 respectively, which one of the following represents the correct order of acid strength? (1)
- A. $\text{C} > \text{A} > \text{B}$ B. $\text{B} > \text{A} > \text{C}$
C. $\text{A} > \text{B} > \text{C}$ D. $\text{C} > \text{B} > \text{A}$
-
67. What is the percent dissociation in 1 M solution of HF containing the concentration of $\text{H}_3\text{O}^+ = 2.7 \times 10^{-2} \text{ M}$? (1)
- A. 37% B. 3.7%
C. 27% D. 2.7%
-
68. The concentration of H_3O^+ and OH^- respectively with $\text{pOH} = 3$ at 25°C are (1)
- A. 1.0×10^{-11} and $1.0 \times 10^{-4} \text{ M}$ B. 1.0×10^{-3} and $1.0 \times 10^{-10} \text{ M}$
C. 1.0×10^{-3} and $1.0 \times 10^{-12} \text{ M}$ D. 1.0×10^{-11} and $1.0 \times 10^{-3} \text{ M}$
-
69. What is the effect if 0.3 M of CH_3COONa is added into aqueous solution of 0.2 M CH_3COOH ? (1)
- A. CH_3COOH becomes more acidic B. The concentration H_3O^+ will increase
C. The equilibrium shifts to the left D. pH value will decrease
-
70. How is a buffer solution prepared? (1)
- A. By mixing a weak acid and a strong acid
B. By mixing a weak acid and its conjugate base.
C. By mixing a strong base and its conjugate acid.
D. By mixing a strong acid and its conjugate base.
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71. Which of the following compounds would be the most basic? (1)
- A. 0.2 M sodium acetate B. 0.2 M acetic acid
C. 0.2 M hydrochloric acid D. 0.2 M ammonium chloride
-
72. What volume of 0.2 M KOH is required to neutralize completely 50 mL of 1.5 M H_2SO_4 ? (1)
- A. 750 mL B. 375 mL
C. 250 mL D. 50 mL
-
73. The pH of the solution for the titration of 100 mL of 0.3 M HNO_3 after addition of 100 mL of 0.3 M KOH is (1)
- A. 12.7 B. 7
C. 2.7 D. 4
-
74. A system that can exchange matter and energy is (1)
- A. closed system B. isolated system
C. open system D. homogeneous system

75. If 1,500 J of heat is added to a system while the system does work equivalent to 2,500 J by expanding against the atmosphere, what is the value of ΔE for the system? (1)

1)

- A. 4,000 J B. 1,000 J
C. -4,000 J D. -1,000 J

— END OF EXAMINATION —